

The Trust Gap in Agentic Search: How Verbal-Imagery Cognitive Styles Shape Behavioural Signals and AI Acceptance

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Abstract—The paradigm of web search is currently shifting from reactive information retrieval to Agentic AI, where proactive systems autonomously synthesise information to assist users. However, for these agents to be effective, they must understand which user parameters drive behaviour to resolve the personalisation cold-start problem. While cognitive architecture is a recognised factor, empirical evidence linking specific traits to web search interaction remains unclear. This paper investigates the Verbal-Imagery (V-I) cognitive style dimension and its influence on proactive search behaviour, mental workload (MWL), and overall search user interface (SUI) alignment. Through a controlled user study ($N = 20$), web search behaviours were evaluated using interaction logs and think-aloud protocols, while MWL and usability were assessed via NASA-TLX and the System Usability Scale (SUS). Our findings reveal that verbalisers and imagers adopt statistically distinct navigational preferences: verbalisers prefer sporadic, reactive interactions, while imagers rely on structured, proactive synthesis such as the *Knowledge Panel*. It is worth noting that qualitative data identifies a “trust gap” in AI-generated overviews based on cognitive modality preferences. These results demonstrate that the V-I dimension is a critical parameter for user modelling in agentic systems. We conclude by proposing requirements for user-aware agentic information retrieval, providing a framework for agents to dynamically adapt their representation strategies to minimise cognitive friction and enhance trust in proactive computing environments.

Index Terms—Cognitive Style; Information Retrieval; Search Engine User Interface Design; Web Search Behaviours; Mental Workload; Trust Gap;

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I. INTRODUCTION

The landscape of web search is currently undergoing a significant paradigm shift from reactive retrieval to Agentic AI (AAI) [1]. Traditionally, search engines have functioned as reactive tools, requiring users to formulate queries and manually filter through results [2]. While these platforms have introduced reactive AI features like *AI Overviews*, they often remain limited by a “one-size-fits-all” representation modality and often rely on traditional AI paradigms that require extensive interaction histories or explicit user modelling [3], [4]. This reliance creates a persistent personalisation cold-start problem: a temporal gap where the interface remains cognitively misaligned with the user until sufficient data is accumulated, often forcing the user to incur unnecessary additional mental workload (MWL) as they adapt to the system [5], [6].

AAI, characterised by its inherent adaptability and self-sufficiency [7], is uniquely positioned to resolve this cold-start problem. Rather than waiting for long-term historical datasets, proactive agents can bridge the personalisation gap by anticipating intent and synthesising information from the very first interaction [8]. To achieve this quick personalisation, agents must be able to decode real-time behavioural signals like navigation patterns and feature engagement. These signals can act as observable proxies for a user’s underlying information-processing habits.

Empirical groundwork is required to decode these signals

and identify the specific user parameters driving them to successfully integrate AAI into web search. Cognitive style offers a particularly robust framework for this purpose, as it dictates an individual's habitual approach to organising and representing information [9], [10]. Because cognitive styles are inherent traits rather than learned behaviours, they influence web search interactions from the onset of a session [11]–[13]. By identifying a user's cognitive profile through initial behavioural cues, an AAI can proactively adjust its representation strategy, effectively bypassing the data-heavy personalising period required by traditional AI personalisation systems.

Research historically categorises cognitive styles into two principal dimensions: the Wholist-Analytic (W-A) and the Verbal-Imagery (V-I) dimensions [14]. These dimensions are distinguished by their conceptual focus: the WA dimension emphasises how individuals organise information, whereas the V-I dimension stresses the modality through which that information is represented. Specifically, individuals with a *wholist* style tend to retain a holistic, global perspective of information, while those with an *analytic* style deconstruct information into its constituent parts for processing [14]–[16]. On the V-I dimension, *verbalisers* prefer representing information through textual and verbal thinking strategies, whereas *imagers* rely primarily on mental pictures and visual representations [14], [15], [17].

While the impact of the WA dimension has been explored in search user interface (SUI) research [12], the V-I dimension remains a critical gap in the development of proactive agents. Previous studies on V-I styles have yielded conflicting conclusions regarding interaction behaviour with different SUI features [11], [18], leaving a critical gap: since users possess inherent traits across both dimensions, overlooking either of them risks creating proactive agents that remain cognitively misaligned with the user from the first interaction.

To address this, our work investigates the empirical requirements for proactive adaptation to resolve the cold-start problem through the following research questions (RQs):

RQ1: How do V-I cognitive styles manifest in user engagement with different SUI features during web search?

RQ2: To what extent does the alignment (or mismatch) between a search system's representation strategy and a user's V-I style impact MWL?

By combining interview data, behavioural patterns, and NASA Task Load Index (NASA-TLX) scores, we analyse how users with different cognitive styles conduct their web search and their unique performance for different SUI features. Our findings provide foundational design requirements for AAI, offering the evidence needed to develop systems that dynamically adapt representation strategies to align with the user's inherent cognitive style.

II. RELATED WORK

A. V-I Cognitive Styles & Web Search Behaviours

By analysing how users with specific cognitive styles interact with web search engines, system providers can design

intuitive SUIs that facilitate efficient information retrieval (IR) [11]. In the context of AAI, behaviours such as strategies, reformulations, and navigation patterns serve as real-time signals that allow an agent to determine how to provide personalised proactive support.

Kinley's model highlights four essential aspects of these behaviours: *Information searching strategies*, *Query reformulation behaviour*, *Web navigation styles*, and *Information processing approaches*. *Information searching strategies* are categorised into *top-down* and *bottom-up* approaches [19]. Individuals with a *top-down* strategy begin with broad concepts and iteratively refine their queries, whereas *bottom-up* users start with specific queries and expand toward general information. Research suggests that verbalisers often exhibit a *top-down* approach, while imagers are more inclined toward a *bottom-up* strategy [11]. *Query reformulation behaviour*, the modification of prior queries to improve results [20], provides a window into a user's cognitive state. For a proactive agent to assist in query expansion effectively, it must anticipate these inherent tendencies. Verbalisers tend to reformulate queries using *add*, *remove*, and *replace* functions more frequently, whereas imagers rely more on *new* and *repeat* queries [11], [21]. *Web navigation styles* describe browsing patterns, categorised as *sporadic* (frequent clicks and tab switching) or *structured* (fewer sites with detailed management) [22]. Verbalisers typically prefer a *sporadic* navigation style, whereas imagers demonstrate a more *structured* approach, often spending more time reading selected pages in detail [11], [22]. Finally, *Information processing approaches*, which refers to how users view and select information, are classified as *scanning*, *reading*, or *mixed* [19]. Verbalisers generally adopt a *scanning* approach via trial and error, while imagers are apt to follow a *reading* strategy.

However, as shown in Table I, findings between studies like Kinley et al. [11] and Hariri et al. [18] remain conflicting. Specifically, participants with the same cognitive style exhibited opposing *Information searching strategies*, *Web navigation styles*, and *Information processing approaches* during web searches. In addition, there is a lack of studies regarding *Query reformulation behaviours* in Hariri et al.'s study [18], leaving a gap in the comparative data. These discrepancies highlight the complex influence of cognitive style and suggest that a “one-size-fits-all” proactive agent may fail to account for these subtle individual differences.

TABLE I
ASSOCIATION BETWEEN V-I COGNITIVE STYLES AND WEB SEARCH BEHAVIOURS IN DIFFERENT STUDIES

Web Search Behaviours	Verbalisers	Imagers
<i>Information searching strategies</i>	Top-down* Bottom-up†	Bottom-up* Top-down†
<i>Query reformulation behaviours</i>	Add, Remove, Replace*	New, Repeat*
<i>Web navigation styles</i>	Sporadic* Structured†	Structured* Mixed†
<i>Information processing approaches</i>	Scanning* Reading†	Reading* Mixed†

*Kinley et al. [11] †Hariri et al. [18]

B. Agentic AI & Information Retrieval

The shift from reactive IR to the agentic web transforms IR from a passive document filtering process into an active, goal-oriented orchestration of tasks [1], [23]. Within this environment, AAI functions as an autonomous *mid-user* that plans, reasons, and executes multi-step workflows to achieve a target information state [1], while human users act as *end-users* who submit high-level requests and expect personalised, completed outputs (see Fig. 1).

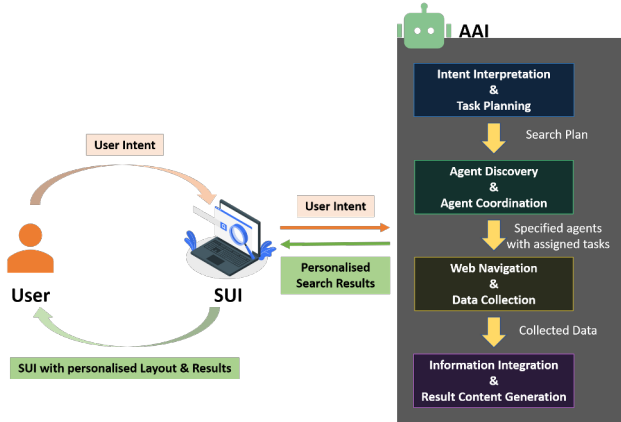


Fig. 1. The cooperation between the users and AAI during IR tasks (adapted from Yang et al. [1])

Although the AAI-user cooperation loop shifts the burden of process-heavy tasks from the users to the agent, the low transparency of the AAI process raises the trust issue regarding the results that AAI agents provide [24], [25]. Especially for non-professional users, the internal reasoning logic and decision-making basis of the agent are difficult to understand, which significantly restricts their ability to assess the rationality and reliability of the agent's output. To address this, Shen [26] proposes a framework for building trustworthy AI. This approach suggests that credibility can be enhanced not by exposing technical implementation details, but by translating the agent's behavioural logic and motivations into a user-understandable format.

In the IR field specifically, as the bridge between users and the system [27], the SUI should be carefully designed to clearly present the logic of the AAI behind the collected results to enhance trust. Aligning with the user's cognitive style is essential, as providing results and layouts that match the user's internal cognitive model can help the user understand the logic, thereby significantly increasing the perceived trust [28], [29]. Despite its potential, research into cognitive-style-matched SUI design for agentic systems remains largely overlooked.

C. Cognitive Styles & Search User Interface Design

Users' cognitive styles fundamentally affect how they perceive and react to user interface (UI) features [30]. Aligning UI design with these styles has been shown to significantly enhance usability across various domains [31], [32].

While SUI designs vary, Wilson [33] provides a robust framework by categorising elements into four primary functional zones: 1) *Input features* for query entry; 2) *Control features* for facet-based refinement and reformulation; 3) *Informational features* for the presentation of results; and 4) *Personalisable features* leveraged from user history. An example is shown in Fig. 2, which adapts this taxonomy to a standard contemporary Google Search¹ SUI.

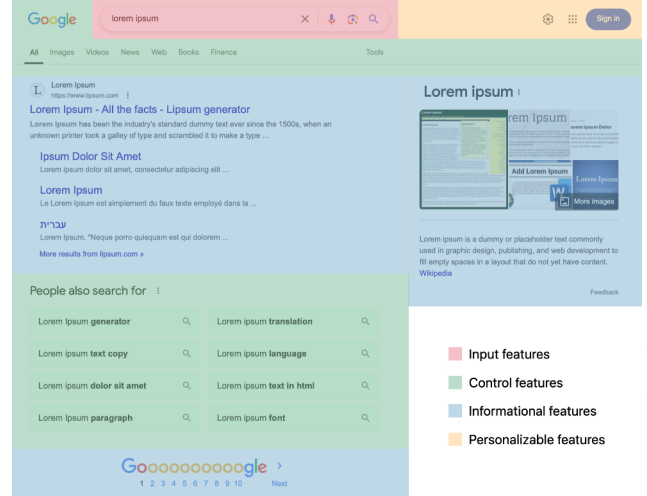


Fig. 2. A contemporary Google SUI, colour-coded to illustrate the four primary feature categories adapted from Wilson's taxonomy [33]: Input (pink), Control (green), Informational (blue), and Personalisable (yellow) features.

In the era of AAI, the SUI is evolving from a passive display into an active workspace. Modern search engines have begun integrating features that shift the IR paradigm toward AAI, such as *AI Overviews* within the *Informational features* zone, or transitioning static results into agent-mediated synthesis and multi-step reasoning via conversational agents like ChatGPT². However, these implementations either remain fundamentally reactive or heavily rely on historical interaction data [3], [4]. The mechanisms for making these features proactively personalised without a cold-start problem, especially based on cognitive styles, remain under-researched.

While the W-A dimension in SUI design is well-documented through examples like wholists preferring broad overviews and analytics favouring nested structures [30] (Fig. 3), the V-I dimension remains critically under-explored in the context of proactive systems. This gap is particularly pressing because proactive features synthesise information both visually and verbally. If an agent's representation strategy is mismatched with the user's V-I style, it risks causing mental overload, where the cognitive cost of processing a non-preferred modality (e.g., an imager being forced to read a long textual synthesis) outweighs the benefit of the proactive intervention. This work investigates how V-I styles influence the perception of these reactive features on the current SUI

¹Google Search – <https://www.google.com>

²ChatGPT – <https://chatgpt.com/>

to establish foundational design requirements for user-aware AAI.



Fig. 3. An example of a flexible UI design that adapts to W-A cognitive styles, proposed by Chen et al. [30]. (a) A compact view with fewer main categories is presented for analytics. (b) For wholists, hovering reveals a horizontal menu with more topics to provide a broader overview.

III. EXPERIMENT DESIGN

A. Hypotheses

This study is guided by two primary hypotheses investigating the influence of the V-I cognitive style on user interactions with the features on the SUI.

H1: V-I cognitive styles will influence how users interact with SUI features during web search.

H2: MWL will vary as an observable function of the alignment (or mismatch) between a user's V-I cognitive style and the SUI's information representation strategy.

The foundation for **H1** lay in findings that individuals approach information retrieval with distinctive strategies, which would influence their usage of different types of SUI features. Building on this, **H2** posits that the cognitive load caused by a mismatch in representation modality will manifest in varied MWL scores [34], [35].

B. Participants

Twenty undergraduate students (9 males, 11 females) from the University of Nottingham Ningbo China were recruited for the study, with a mean age of 21.94 years. Inclusion criteria required that all participants had normal or corrected-to-normal vision and were proficient in conducting English-language Google searches for unfamiliar topics ($M_{Google_familiarity} = 3.71/5$). The study protocol received full ethical approval from the Faculty Ethics Committee (Approval Number: FOSE-202425-056). Before participating, all individuals received a detailed information sheet and signed written informed consent. Participants were also explicitly informed that their participation was voluntary and that they could withdraw at any time without consequence.

To minimise the influence of prior domain knowledge on search behaviour [36], we assigned search tasks based on a pre-study familiarity questionnaire. Each participant was assigned the two topics with which they were least familiar, thereby minimising pre-existing knowledge as a confounding variable in their search strategy.

C. Apparatus

The user study was conducted in a controlled laboratory environment, configured to resemble a standard office workstation to ensure a high degree of ecological validity (see Fig. 4). Each participant used a dedicated workstation

equipped with a monitor, keyboard, and mouse. The researcher was seated nearby, positioned to observe without causing interference. Search tasks were performed using Google Chrome (v.134.0.6998.183), selected for its widespread familiarity [37] to minimise usability-related confounds. The browser's region was localised to Ningbo to maintain consistency in the search results' resources/rankings and AI Overviews across the entire participant cohort.

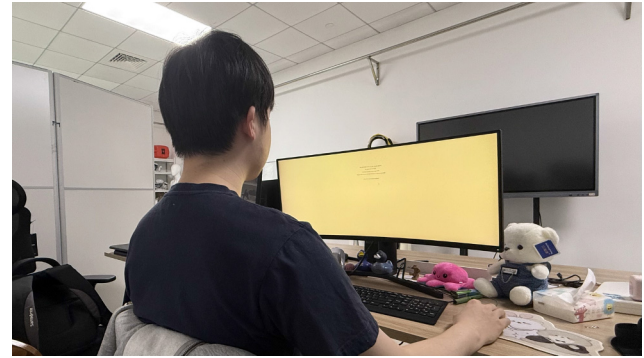


Fig. 4. The experimental setup, designed to simulate a typical office environment to enhance the ecological validity of the study.

D. Web Search Tasks

To ensure the results are generalisable for all types of web search tasks, two tasks were designed to trigger different levels of interaction with SUI features.

- **Lookup Task** – Assume that a friend asks you about [Object-Oriented Programming Principles/Gestalt Principles/Six-Sigma Methodology]. Please write a single, concise sentence that clearly defines the topic. Additionally, provide 3–5 bullet points outlining the key concepts.

Rationale: This task aims to see if users rely on the *Featured Snippet* or *AI Overview* (the reactive “Agent”) to satisfy a quick information need or if they still prefer to browse source pages.

- **Exploratory Task** – Assume that you have just received a video interview invitation from your dream company. Please write a brief paragraph outlining how you would prepare for it.

Rationale: This task requires open-ended discovery, testing how users interact with *Related Questions* as agentic guides that help navigate a complex information space.

E. Data Collection

To address the hypotheses, we collected three primary types of data: behavioural metrics, subjective MWL ratings, and qualitative user experience feedback.

a) *Web Search Behavioural Data:* Guided by the web search behaviour framework from Section II-A, we logged and analysed participants' quantitative interactions with the search engine. These metrics related to *Information searching strategies* and *Query reformulation behaviours*, assessed by

analysing how users formulated new queries or modified existing ones. Furthermore, we evaluated *Web navigational styles* and *Information processing approaches* based on screen recordings, with a focus on metrics such as cumulative clicks, the number of concurrently open web pages, and page dwell time, to classify the behaviours into *sporadic* or *structured* web navigation styles, and whether they are *scanning* or *reading* the opened pages. To ensure reliability, these behaviours were coded and verified by two reviewers. Qualitative insights into their preferences for how they perceive and process search results were gathered from think-aloud protocols (described below) and debriefing interviews.

b) Subjective Mental Workload: Rather than being inherently negative, elevated MWL can indicate deep engagement with search results or, conversely, difficulty in locating needed information [38]. To assess this, we administered NASA-TLX during the debriefing session. The NASA-TLX is a widely validated multidimensional scale that assesses perceived workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration level [39]. It was selected over other assessment tools for its suitability for post-task evaluation [40] and its non-intrusive nature, which avoids the potential confounds of real-time assessments like the Instantaneous Self-Assessment (ISA) [41].

c) Subjective User Experience: Participants' qualitative feedback on specific SUI features was crucial for investigating their design preferences. We collected this feedback through two methods: spontaneous verbal expressions during the think-aloud protocol and direct questioning during the post-task debriefing session. The investigation focused on the six SUI features illustrated in Fig. 5.

In addition, we administered the System Usability Scale (SUS) questionnaire [42] after the task to complement the qualitative data. The SUS is a reliable, ten-item tool for assessing a system's overall usability, efficiency, and effectiveness, yielding a score out of 100 [43], [44]. We incorporated it to provide a holistic, quantitative measure of user satisfaction with the whole SUI.

F. Procedure

The study was conducted in English and divided into two sittings to separate the cognitive style assessment from the web search tasks. This bifurcation allowed sufficient time to process the cognitive style data and ensure a balanced recruitment of participants from across the V-I spectrum.

In the first sitting, participants were introduced to the study's objectives and gave informed consent. We then measured their cognitive style along the V-I dimension using a computer-based Verbal Imagery Cognitive Style (VICS) test [17]. This instrument was chosen as it is a reliable tool specifically designed for this purpose, addressing some limitations of the Cognitive Styles Analysis (CSA) [45] on V-I measurements. The VICS test consists of two sessions. In the verbal session, participants were asked to determine whether the two items shown on the screen are both *natural*, both *man-made*, or a



Fig. 5. The six core SUI features investigated in this study, shown on a typical Google search results page: (1) Toolbar, (2) Featured Snippet/AI Overview, (3) Basic Snippets, (4) Knowledge Panel, (5) Related Questions ('People also ask'), and (6) Related Searches.

mixed pair (one natural and one man-made). In the imagery session, they were required to compare the relative real-world sizes of the two items. Cognitive style was determined by a ratio of the median reaction time to the verbal session and that to the imagery session (V-I ratio). The cutoff point for cognitive style classification is chosen as the mean V-I ratio if the data is normally distributed, and the median V-I ratio otherwise [46], [47]. The VICS test took approximately 30 minutes to complete, after which the participants were dismissed and informed that they would receive instructions for the second sitting via email. All the participants were then classified as verbalisers or imagers based on the mean V-I ratio ($M = 1.097, SD = 0.14$) after the normal distribution was checked by a Shapiro-Wilk test [48] ($p = 0.08 > 0.05$).

The second sitting focused on the web search tasks. Throughout these tasks, we employed an Invasive Think-Aloud Protocol (ITAP) [49]. Crucially, the researcher used ITAP to probe the user's trust and reliance on reactive AI features (e.g., asking "Why did you choose to read the AI Overview instead of clicking the first link?") during the search task to elicit deep insights into user mental models. This data was later transcribed verbatim for analysis. The search tasks lasted around 40 minutes, depending on individual progress.

Immediately after completing the search tasks, participants completed two questionnaires: the SUS and the NASA-TLX to assess the interface usability and their subjective MWL, respectively. Following this, a semi-structured debriefing in-

terview was conducted to summarise their overall experience, discuss the SUI features they found helpful or redundant, and explore the reasoning behind their search strategies.

IV. RESULTS

This section presents the statistical analyses of user interactions with search features. Data from the lookup and exploratory tasks were synthesised and analysed as a single, continuous search session. This approach reflects the fluid nature of real-world information searching, where users frequently transition between goal-oriented retrieval and open-ended discovery. For an AAI to successfully resolve the personalisation cold-start problem, it must be able to identify consistent behavioural signals within these mixed-intent environments without requiring explicit task labelling. A significance level of $p < 0.05$ was used for all tests, supplemented by effect size measures (Cohen's d for t -tests and Cramer's V for Chi-square) to indicate the magnitude of the findings.

Objective task performance, such as response accuracy, completion time, or search precision, was intentionally excluded from this study for three primary reasons:

- **Process-Oriented Focus:** The investigation aimed to identify latent behavioural signals and cognitive transitions (e.g., *sporadic* vs. *structured* navigation) rather than the raw utility of the search engine. Since the goal is to inform the design of user-aware proactive agents, the process of inquiry is more informative than the quality of the output, which frequently reflects the search engine's ranking algorithms rather than the user's cognitive strategy.
- **Ecological Validity:** Omitting performance pressure preserved the ecological validity of the study. This allowed participants to interact with *AI Overviews* and agentic features naturally, avoiding the "evaluation anxiety" that often triggers artificial, hyper-cautious search behaviours.
- **Integration of ITAP:** The use of the ITAP made task duration influenced not only by search efficiency but also by the frequency and depth of concurrent participant-researcher dialogue, which varied between participants and sessions based on their unique interactions.

Instead of objective metrics, the effectiveness of the interaction was evaluated through a combination of the SUS and semi-structured debriefing interviews. This approach facilitated a holistic assessment of whether the SUI supported participants' information needs while capturing feedback on specific features relative to their cognitive styles. In addition, perceived success was captured via the *Performance* subscale of the NASA-TLX, providing a subjective measure of achievement that reflects the user's internal sense of task completion and efficiency.

A. Web Search Behaviours as Agentic Signals (H1)

To identify behavioural signals that could inform a user-aware proactive agent, we analysed the association between the V-I dimension and session-wide search behaviours using Chi-square tests [50]. The results are summarised in Fig. 6.

The analysis revealed a statistically significant association between the V-I dimension and participants' *Web navigation style* ($\chi^2(1) = 4.85, p = 0.03$). The strength of this association was found to be of a medium-to-large effect size (Cramer's $V = 0.49$). verbalisers exhibited a strong preference for *sporadic navigation* (81.8%), whereas imagers predominantly adopted a *structured style* (66.7%), which aligned with Kinley et al. [11]'s findings but conflicted with Hariri et al. [18]'s, who found the exact opposite preference. This consistency across mixed tasks suggests that navigation style is a promising "cognitive signal" that an agent can detect early in a session. No other statistically significant associations were found for *Information searching strategy* ($\chi^2(1) = 1.62, p = 0.20$), *Query reformulation behaviours* (Table II), or *Information processing approaches* ($\chi^2(2) = 4.34, p = 0.11$).

TABLE II
CHI-SQUARE TEST RESULTS FOR QUERY REFORMULATION BEHAVIOURS BY COGNITIVE STYLE.

Statistic Result	Add	Replace	Repeat	New	Remove
$\chi^2(p)$	0.15(0.70)	0.28(0.60)	-	2.04(0.51)	1.02(0.31)
Cramer's V	0.09	0.12	-	0.33	0.23

B. Perception of Agentic SUI Features (H1)

a) **Overall Usability (SUS Scores):** Data normality for SUS scores was confirmed (Shapiro-Wilk [48] $p = 0.37 > 0.05$). An independent-samples t -test [51] revealed a significant difference in *Usability* ratings ($t(18) = 2.41, p = 0.03$). Imagers perceived the overall search interface as significantly more usable ($M = 85.28, SD = 4.91$) than verbalisers ($M = 78.41, SD = 7.27$). The calculated effect size was large (Cohen's $d = 1.08$), indicating that the current SUI provides a structurally superior experience for imagery-oriented users while posing a significant usability barrier for verbalisers.

b) **Feature Preference and Requirements:** Chi-square tests [50] (reported as χ^2 in Table III) analysed the relationship between V-I styles and preferences for six specific features. The features that showed significant differences between V-I participants have been shown in Table III.

TABLE III
SUI FEATURE PREFERENCE FREQUENCY BY V-I COGNITIVE STYLE

Feature	verbalisers ($n = 11$)	Imagers ($n = 9$)	$\chi^2(p)$	Cramer's V
Knowledge Panel	2 (18.2%)	8 (88.9%)	7.27 (0.007)	0.70
Toolbar	9 (81.8%)	1 (11.1%)	4.90 (0.03)	0.60

Combined with the ITAP and debriefing interview data (qualitative data), three features in total showed noticeable differences:

- **Knowledge Panel:** A highly significant association was found ($V = 0.70$). Imagers preferred this agentic feature for its visual convenience. However, Participant 7 (imager) noted: "Sometimes the panel is full of texts, making it hard for me to identify which part is useful", suggesting a requirement for modality-adaptive synthesis.
- **Toolbar:** A significant association was also found ($V = 0.60$). verbalisers utilised the toolbar as a high-utility

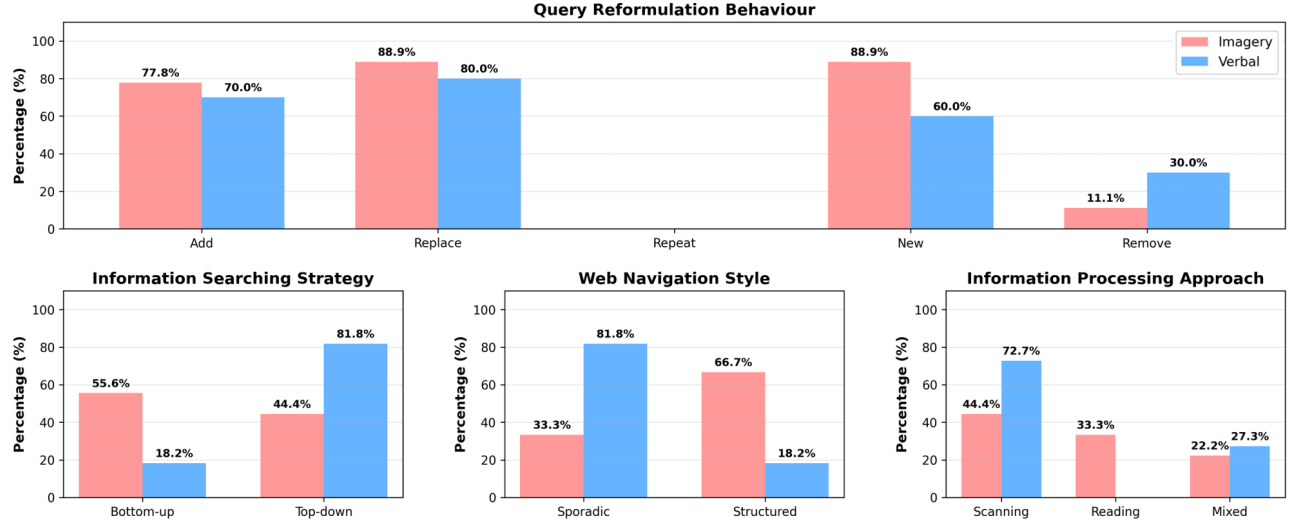


Fig. 6. Distribution of web search behaviours categorised by V-I cognitive style. The axes represent the percentage of participants within each style group (Y-axis) relative to specific behavioural categories (X-axis).

control feature for active filtering, whereas some of the imagers viewed it as “redundant”.

- **AI Overview/Featured Snippets:** While statistical significance was not reached for preference alone, qualitative data revealed a distinct “**trust gap**”. The majority of verbalisers (91% (10/11)) praised the feature for providing a “quick answer”, whereas 67% (6/9) of imagers expressed scepticism regarding agentic results, perceiving them as “insufficiently attributed”.

These results support *H1*, demonstrating that user cognitive styles dictate fundamental requirements for trust, modality, and control in emerging agentic search environments.

C. Mental Workload on SUIs (H2)

Subjective MWL, as measured by the NASA-TLX scores, were normally distributed as checked by a Shapiro-Wilk test [48] ($p = 0.39 > 0.05$). To evaluate *H2*, independent-samples *t*-tests [51] were conducted to determine if current search interfaces impose disproportionate cognitive demands on specific V-I profiles. The analysis revealed no significant effect of cognitive style on overall MWL scores ($t(18) = -0.45$, $p = 0.66$), with verbal users ($M = 44.70$, $SD = 10.67$) and imagery users ($M = 47.56$, $SD = 17.67$) reporting comparable levels of cognitive effort. A further examination of the six NASA-TLX subscales (*Mental Demand*, *Physical Demand*, *Temporal Demand*, *Performance*, *Effort*, and *Frustration*) also yielded no significant associations with cognitive style (all $p > 0.05$) (see Fig. 7). Notably, although not reaching statistical significance, verbalisers reported marginally higher weighted *Effort* scores ($M = 123.22$, $SD = 69.82$) compared to imagers ($M = 79.48$, $SD = 35.96$), with a difference of 43.74 points approaching significance ($t(18) = -1.70$, $p = 0.11$).

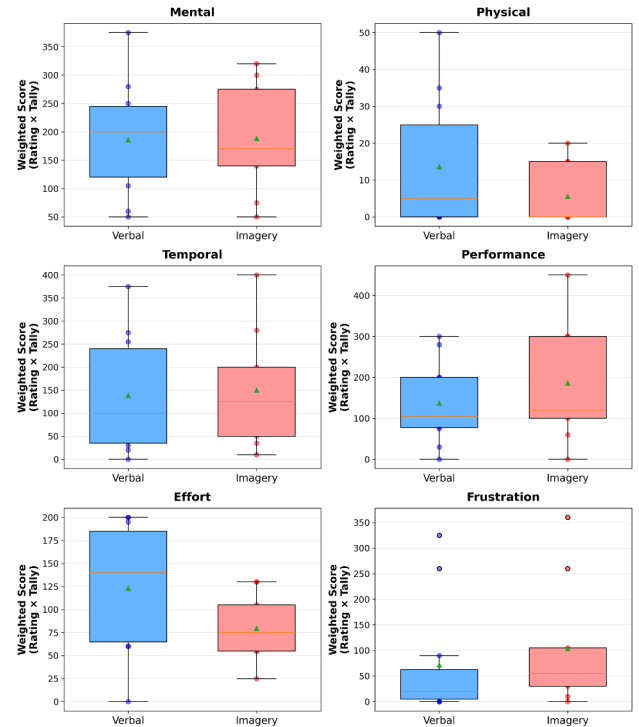


Fig. 7. NASA-TLX Weighted Subscale Scores by Cognitive Style

V. DISCUSSION

The results provide empirical evidence that cognitive style serves as a stable, observable parameter for transitioning from reactive retrieval to AAI. By analysing search behaviours in a mixed-intent environment, we provide preliminary support that

certain behavioural signals are detectable regardless of task complexity, offering a foundation for informing the design of user-aware proactive agents.

A. Main Findings: Foundations for Agentic Modelling

a) *H1: Navigational Patterns as Agentic Feature Design Basis*: Our findings confirm that navigation style is a statistically significant signal ($p = 0.03$) for cognitive style inference. The preference of verbalisers for *sporadic navigation* suggests a “breadth-first” approach, where users collect textual fragments from diverse sources. In contrast, imagers adopt a structured, holistic approach.

These patterns can be used for predictive signals for an AAI. As they remained consistent across pooled lookup and exploratory tasks, an agent can theoretically instantiate a personalised SUI within the first few interactions of a session.

The high utility of the *Toolbar* for verbalisers and the *Knowledge Panel* for imagers further defines the target for this adaptation: verbalisers require high-utility control features to manage their sporadic discovery, while imagers require synthesised, spatial representations.

However, the qualitative “trust gap” identified in *AI Overviews* suggests that for an agent to be successfully proactive, it must not only match the user’s format but also provide transparent attribution to satisfy the verification strategies of imagery-oriented users.

b) *H2: The Importance of V-I Cognitive Style Matched SUI Design*: The lack of a significant difference in NASA-TLX scores ($p = 0.66$) indicates that current SUIs do not yet impose a different MWL on users with different cognitive styles. This may be attributed to “interface familiarity”, where participants have adapted to the standard SUI of modern search. However, the differences in the sub-scales still highlighted the potential impact of the current SUI designs on different V-I cognitive style participants.

Although not reaching statistical significance ($p = 0.11$), the higher *Effort* reported by verbalisers suggests they may have expended greater cognitive effort to achieve comparable task performance. While this trend should be treated as exploratory, given the current sample size, it aligns directionally with the SUS results, where verbalisers provided significantly lower *Usability* ratings. Taken together, these findings suggest that current SUI layouts, despite being familiar, remain a potential source of structural cognitive friction for these users.

B. Implications for Proactive Agentic Systems

To move beyond reactive search, we propose Agentic Adaptation Requirements (AARs). These requirements detected the user’s style via real-time behavioural signals identified in Section IV-A to resolve the personalisation cold-start problem, allowing the agent to align its representation modality with the user’s cognitive style during the initial interaction.

a) AAR1: Requirements for Imagers:

[AAR 1.1] Upon detecting structured navigation, the agent should proactively expand structured visual summaries beyond the standard interface. *Knowledge*

Panels can be seen as an example, towards which imagers expressed a significant preference. This supports a holistic overview and reduces the need for manual tab management.

[AAR 1.2] To bridge the identified “trust gap” towards AI-generated summaries observed in qualitative data, agentic components should provide more source attribution alongside their outputs. As this is drawn from qualitative data rather than a statistical test, it should be treated as an exploratory finding that requires further validation.

b) AAR2: Requirements for Verbalisers:

[AAR 2.1] For users exhibiting *sporadic navigation*, agents should prioritise key information throughout the interface, such as through bold text highlights or concise direct answers, to minimise the effort of scanning across multiple sources. The verbalisers’ perception of *AI Overviews* as a quick-answer feature suggests that such formats can serve as a useful design reference for this adaptation.

[AAR 2.2] To preserve the user’s preference for active, manual discovery, the agent should provide “Control Features” such as advanced filters and query reformulations, rather than automating the entire search flow. This is grounded in the significant preference for the *Toolbar*, which verbalisers consistently utilised as a high-utility mechanism for actively controlling and filtering their search process.

C. Limitations & Future Work

While this study establishes a foundation for user-aware agents, subsequent research should prioritise larger and more diverse cohorts that encompass a broader range of professional and demographic backgrounds to enhance the generalisability of the findings. The sample size of $N=20$ also limits the statistical power of the study, meaning that some null results, particularly for search behaviours and NASA-TLX subscales, may reflect insufficient statistical power to detect subtle behavioural differences rather than the genuine absence of cognitive-style effects. In addition, future work should incorporate a wider variety and a higher volume of search tasks. Expanding the diversity and quantity of tasks will enable a more granular observation of potential differences in *Information searching strategies*, *Information processing approaches*, and *Query reformulation patterns*, ultimately leading to more robust models of user behaviour in agentic environments. Future work should also include objective performance metrics alongside behavioural analyses to evaluate whether style-aware adaptations improve both user experience and search effectiveness.

The next frontier for this research is the implementation of a real-time inference engine within an AAI IR architecture. Future work will focus on evaluating how quickly an agent can switch from a generic SUI to a style-congruent proactive SUI that adapts interface structure, modality, and control mechanisms to users’ cognitive preferences based on

the findings in this work. By further validating these AARs in a live environment, we can move closer to search systems that don't just find the right answer, but present it in the right mind-space.

VI. CONCLUSION

This paper has investigated the influence of the Verbal-Imagery (V-I) cognitive dimension on web search behaviours and user perceptions of search user interface (SUI) features, framing these interactions as foundational requirements for the era of Agentic AI (AAI). Our work specifically addresses the personalisation cold-start problem, defined as the gap where user experience suffers due to a lack of historical data, by proposing cognitive style as a stable and early-identifiable predictor of user needs.

Our findings demonstrate that the V-I cognitive style is a significant factor in how users navigate and process information within web search, and the SUI features that conflict with a user's cognitive profile could cause trust issues. Specifically, we observed distinct navigational "signatures": verbalisers adopted a *sporadic* style, whereas imagers employed a *structured* approach to construct detailed overviews. These behaviours were also reflected in divergent preferences for SUI features: while verbalisers prioritised the control of the *Toolbar*, imagers significantly favoured the reactive synthesis provided by the *Knowledge Panel*, but expressed concern about the reliability of the *AI Overview* feature.

By identifying these cognitive-style-dependent patterns, we provide the evidence base for the user-aware agentic information retrieval. In this paradigm, agents do not merely react to queries but autonomously adapt their representation strategies, choosing between sequential textual scaffolding or holistic visual synthesis, based on the inferred cognitive profile of the user. As we move from reactive search models to proactive systems, accommodating this cognitive diversity is not merely a usability concern but a technical necessity for ensuring system transparency, trust, and alignment in the AAI landscape.

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